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# Analog Electronics lab session

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# Summary

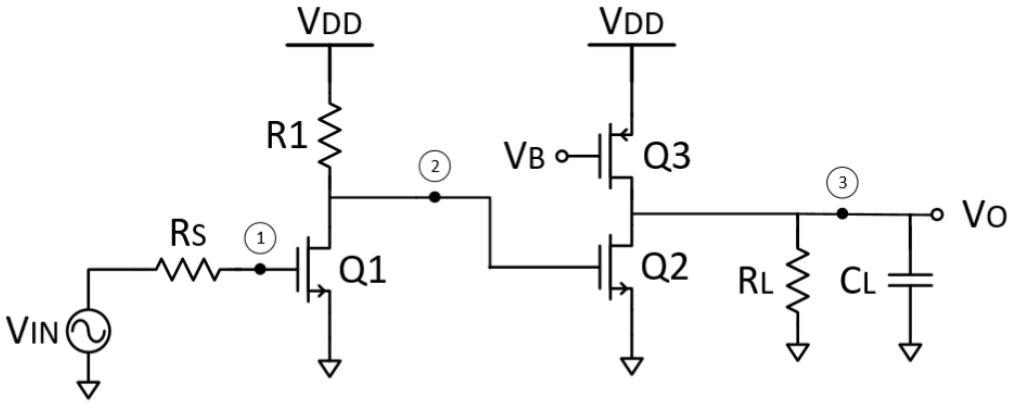
- Session 2:
  - Cascade of common source amplifier stages
  - The two-stage amplifier with Miller compensation

# Building some intuition

- Do exercise 1
  - ETA: 25 minutes
  - Hints: Neglect the smaller capacitance introduced by Miller effect

# EXERCISE 1-A

## Finding the Total Gain



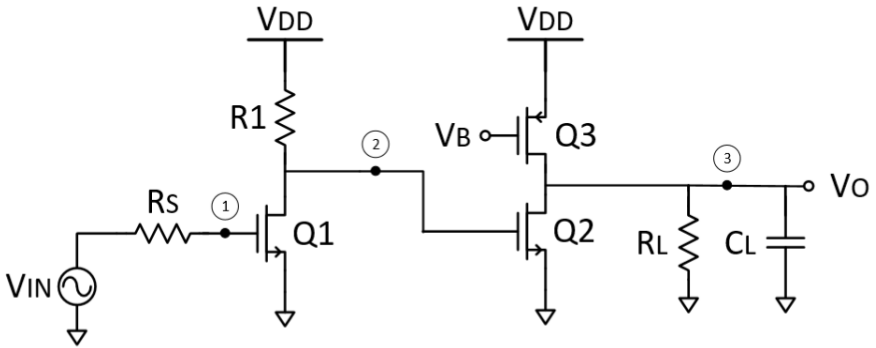

- Gain= $\frac{V_0}{V_{in}} = \frac{I_0}{V_{in}} * \frac{V_0}{I_0} = G_m * R_{out}$

- $A_{v1} = -g_{m1} * R_1 = -2mS * 5k\Omega = -10V/V$
- $A_{v2} = -g_{m2} * R_L = -20mS * 500\Omega = -10V/V$
- $A_v = A_{v1} * A_{v2} = 100\,V/V = 40dB$

# EXERCISE 1-B

## Finding Pole Frequencies

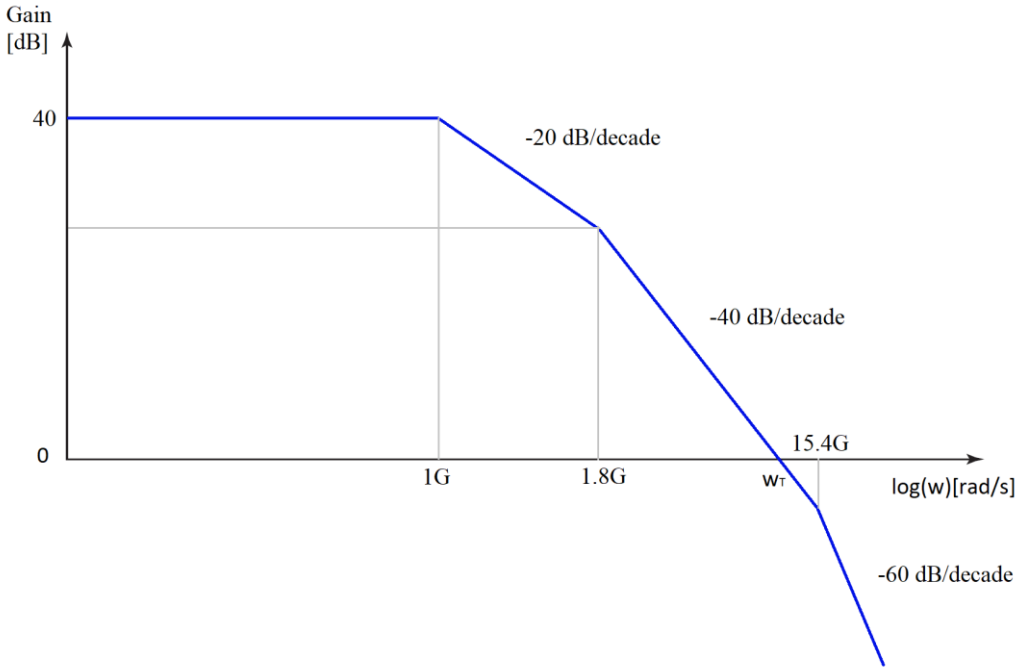
- $pole\ frequency\ at\ node\ n\ in\ rad/s = \frac{1}{Total\ resistance\ between\ node\ n\ and\ AC\ ground} * \frac{1}{Total\ capacitance\ between\ node\ n\ and\ AC\ ground}$




- $\omega_{p1} = \frac{1}{R_S*(C_{gs1}+C_{gd1}*(1+g_{m1}*R_1))} = \frac{1}{500\Omega*130fF} = 15.4G\ rad/s$
- $\omega_{p2} = \frac{1}{R_1*(C_{gs2}+C_{gd2}*(1+g_{m2}*R_L))} = \frac{1}{5k\Omega*195fF} = 1G\ rad/s$
- $\omega_{p3} = \frac{1}{R_L*C_L} = \frac{1}{500\Omega*1.1pF} = 1.8G\ rad/s$

# EXERCISE 1-C

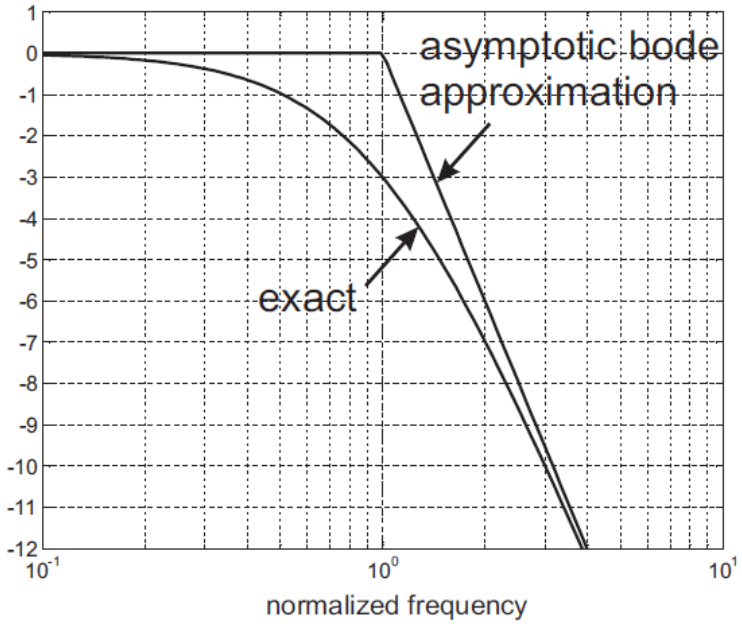
## Finding $\omega_T$ Value




- $40 - 20 * \log\left(\frac{1.8G\ rad/s}{1Grad/s}\right) = 34.9dB$
- $34.9 - 40 * \log\left(\frac{\omega_T}{1.8Grad/s}\right) = 0dB$
- $\omega_T = 13.4G$ rad/s

# EXERCISE 1-C

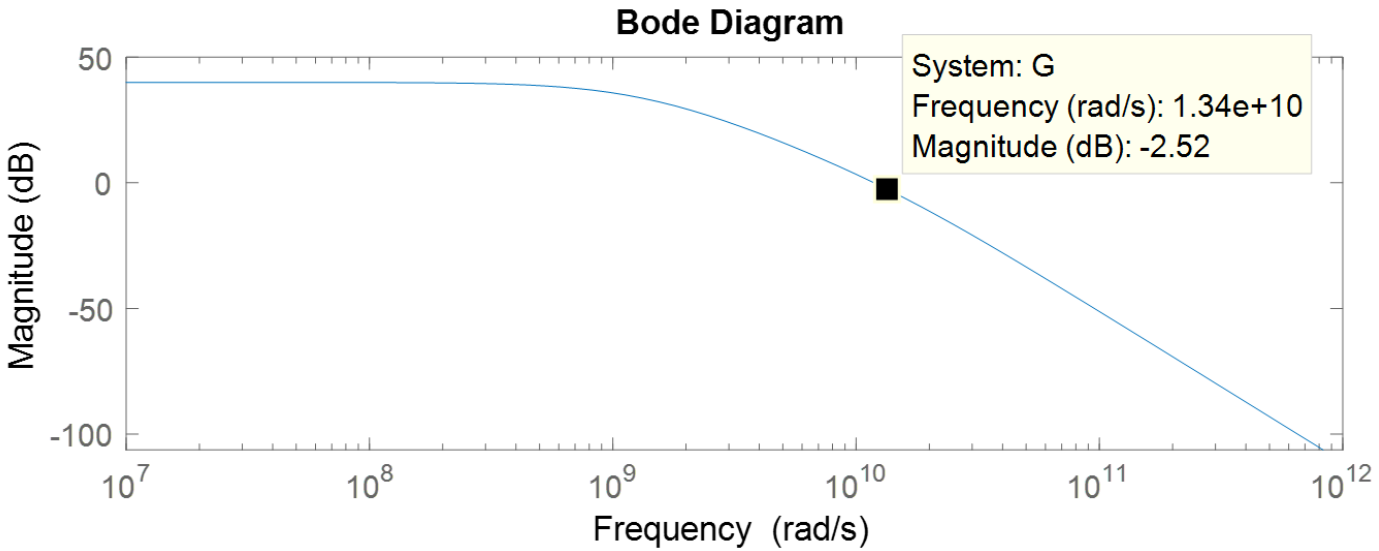
## Asymptotic Bode Approximation vs Exact Curve



# EXERCISE 1-C

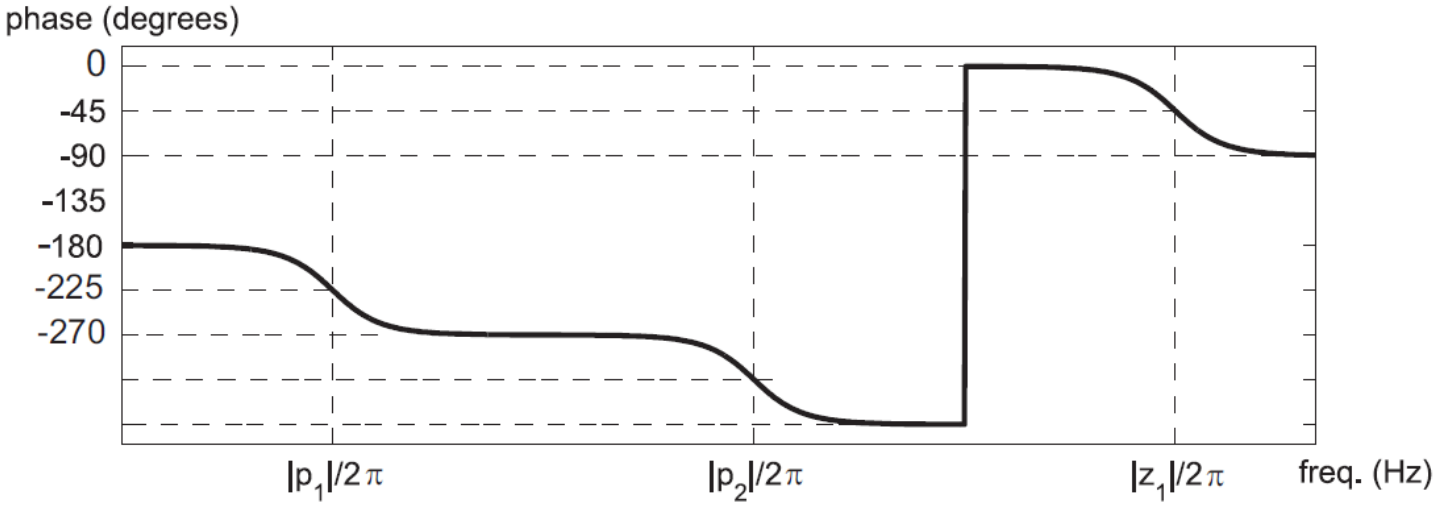
## MATLAB Magnitude Plot

- Calculated $\omega_T = 13.4G\ rad/s$



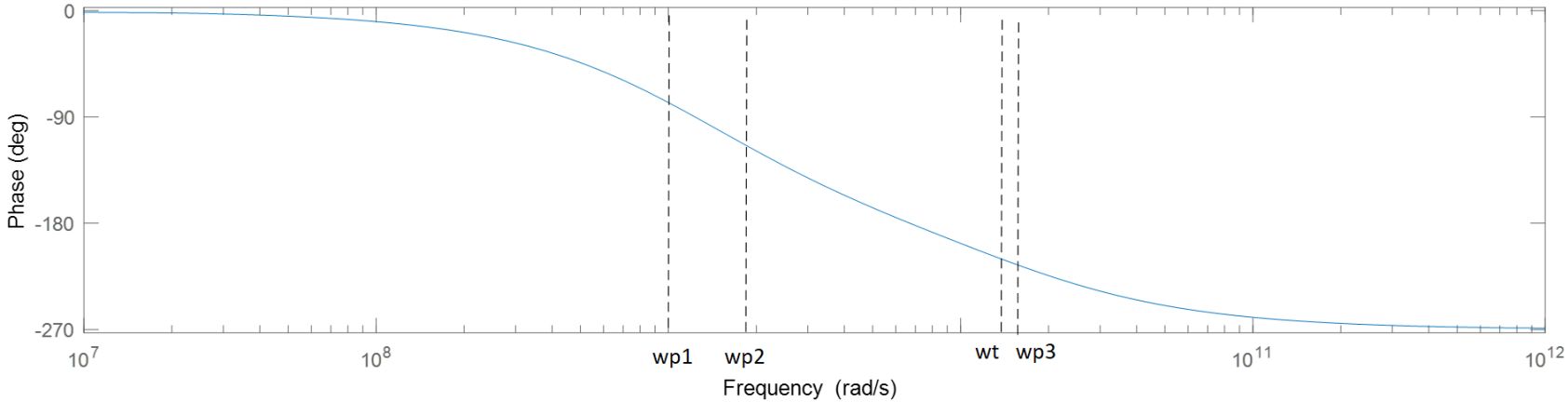
# EXERCISE 1-C

## Phase Shift due to Poles and Zeros



# EXERCISE 1-C

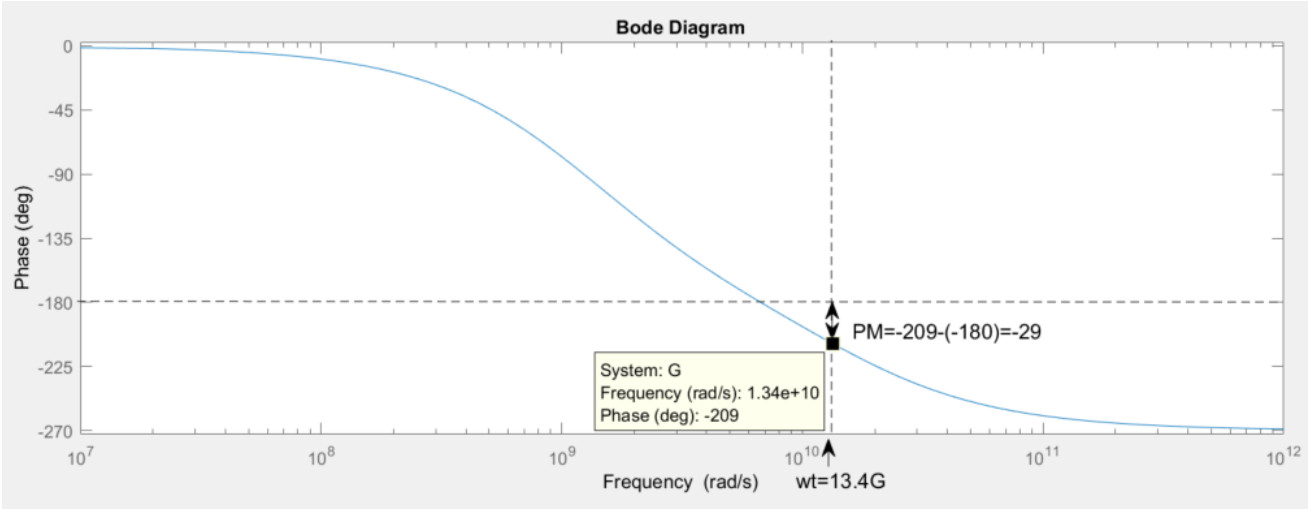
## MATLAB Phase Plot



# EXERCISE 1-C

## Finding Phase Margin

- $Total\ phase\ shift = 0 - tan^{-1}\frac{13.4Grad/s}{1Grad/s} - tan^{-1}\frac{13.4Grad/s}{1.8Grad/s} - tan^{-1}\frac{13.4Grad/s}{15.4Grad/s} = -209.1^o$
- $Phase\ margin = -209.1^o - (-180^o) = -29.1^o$

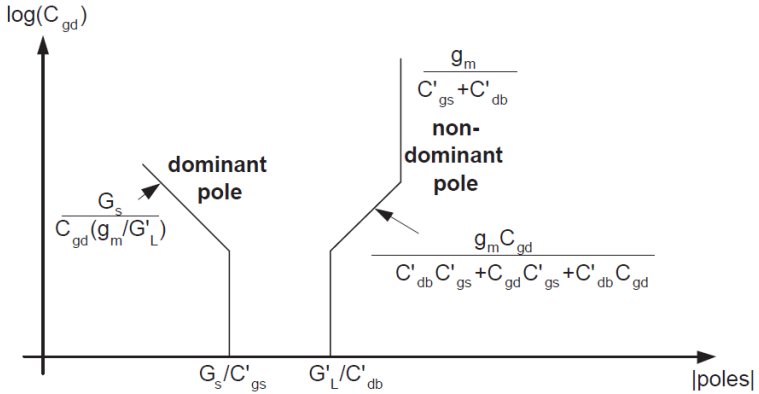


# Building some intuition

- Do exercise 2
  - ETA: 25 minutes

# EXERCISE 2-A

## Dependence of Poles on $C_{gd}$ value



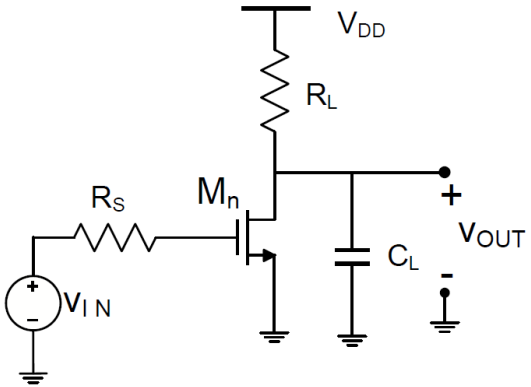
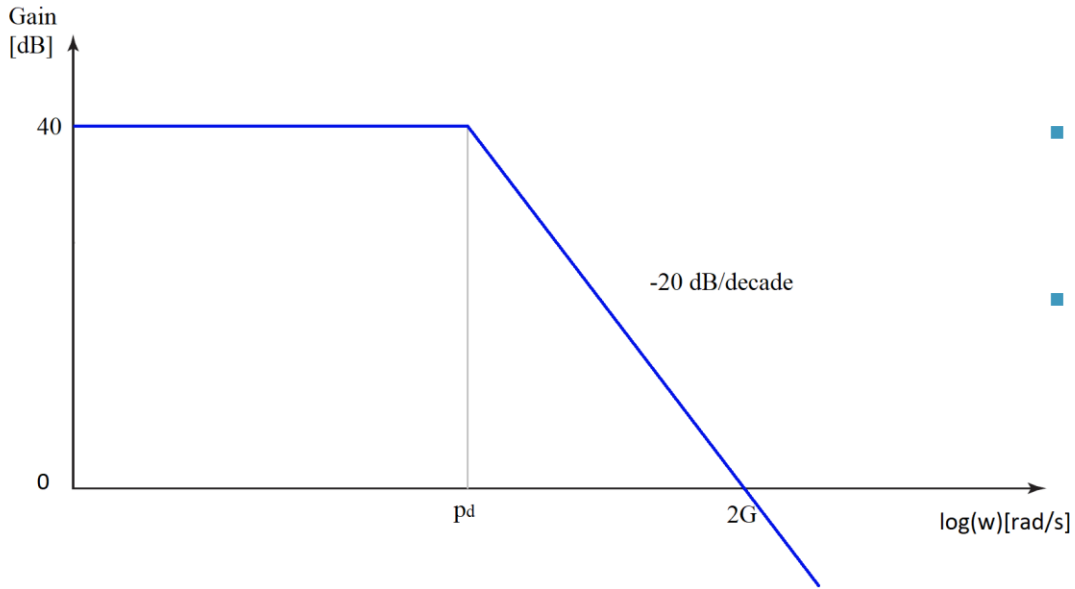




*Figure 2.15: Bode diagram of pole splitting: asymptotes indicate the position of poles as a function of $C_{gd}$ on a log-log scale.*

- $C'_{gs} = C_{gs} + C_{gb}$
- $C'_{db} = C_{db} + C_L$
- $G'_L = g_{ds} + G_L$

# EXERCISE 2-A

## Finding the Value of $C_c$



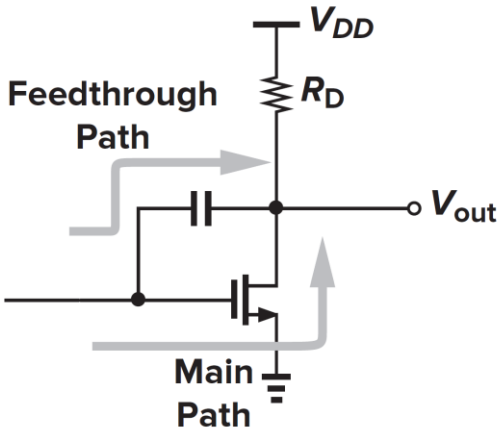

- $40 = 20 * \log\left(\frac{\omega_T}{\omega_{p2}}\right)$

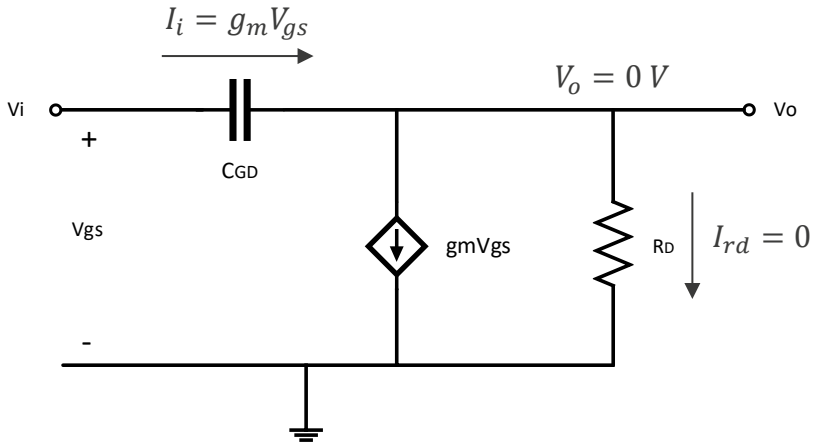
  => $\omega_{p2} = 20Mrad/s$

- $|p_d| = \omega_{p2} = \frac{1}{R_1 * (C_c * g_{m2} * R_L)}$

  => $C_c = 1pF$

# DERIVING REAL ZEROS

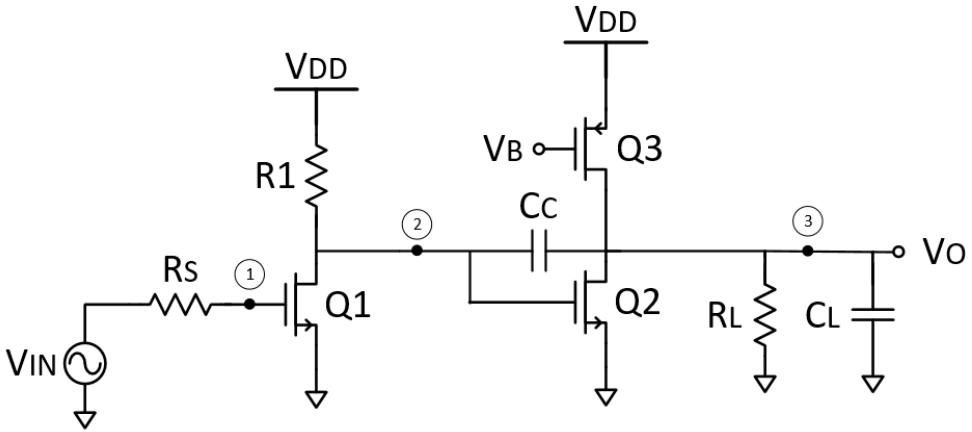



- $\frac{V_i}{1/(s_z * C_{GD})} = g_m * V_i$
- $s_z = \frac{g_m}{C_{GD}}$

# EXERCISE 2-B

## Finding Pole Frequencies

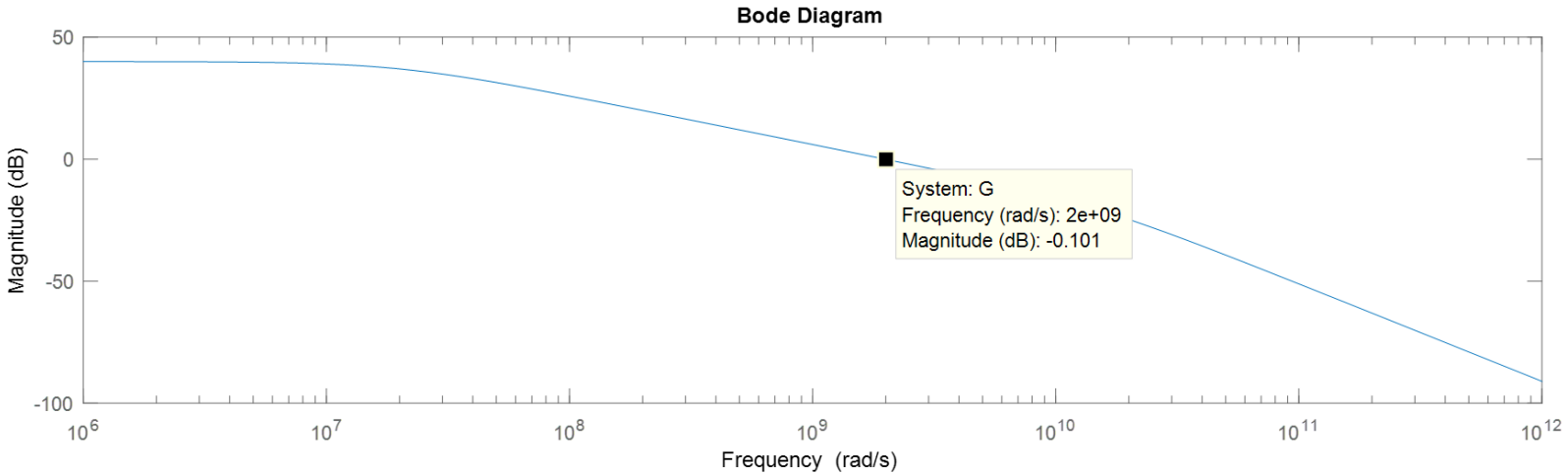



- $\omega_{p1} = \frac{1}{R_S*(C_{gs1}+C_{gd1}*(1+g_{m1}*R_1))} = \frac{1}{500\Omega*130fF} = 15.4G\ rad/s$
- $\omega_{p2} = \frac{1}{R_1*(C_c*g_{m2}*R_L)} = \frac{1}{5k\Omega*20mS*500\Omega*1pF} = 20M\ rad/s$
- $\omega_{p3} = \frac{g_{m2}}{C_L} = \frac{20mS}{1.1pF} = 18.2G\ rad/s$
- $\omega_{z1} = \frac{g_{m2}}{C_c} = \frac{20mS}{1pF} = 20G\ rad/s$

# EXERCISE 2-B

## MATLAB Magnitude Plot

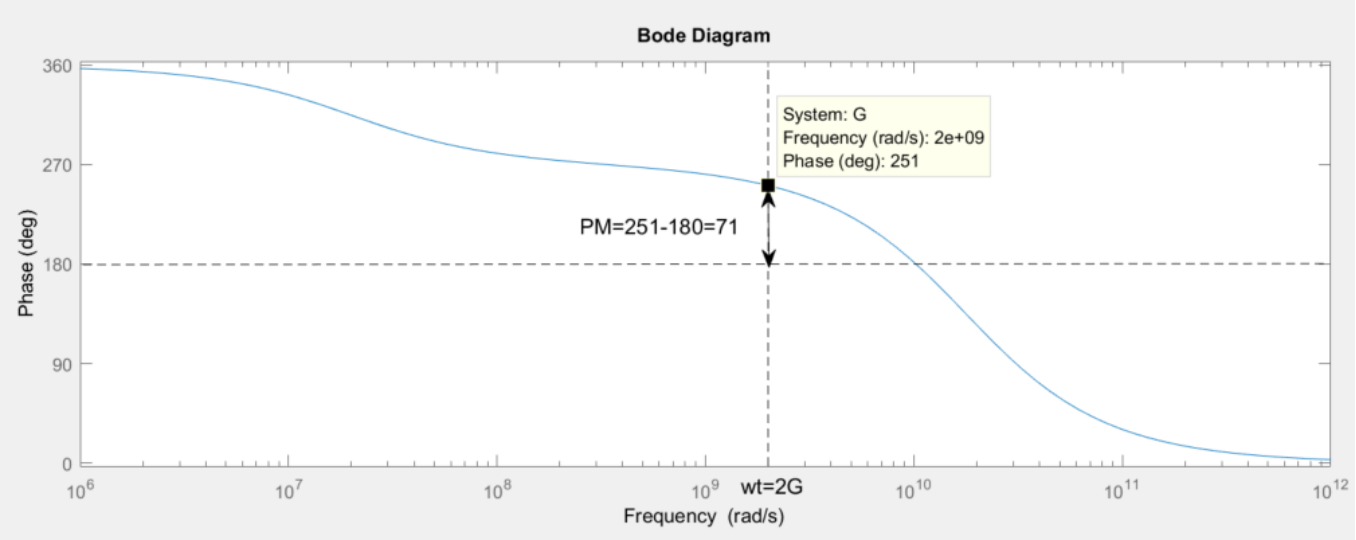
- Calculated $\omega_T = 2G\ rad/s$



# EXERCISE 2-B

## Finding Phase Margin

- $Total\ phase\ shift = 0 - tan^{-1}\frac{2Grad/s}{20Mrad/s} - tan^{-1}\frac{2Grad/s}{15.4Grad/s} - tan^{-1}\frac{2Grad/s}{18.2Grad/s} - tan^{-1}\frac{2Grad/s}{20Grad/s} = -108.8^o$
- $Phase\ margin = -108.8^o - (-180^o) = 71.2^o$

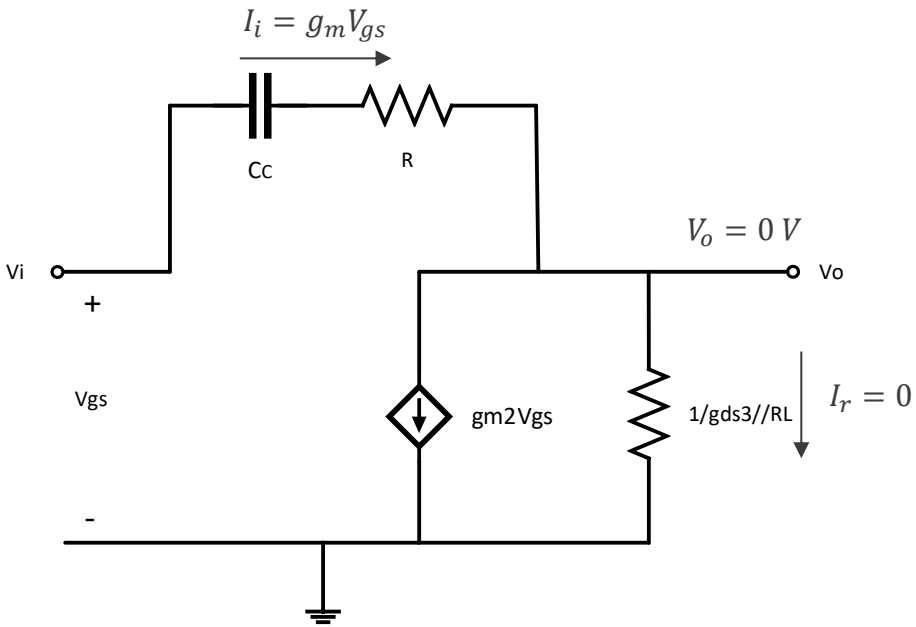


# Building some intuition

- Do exercise 3
  - ETA: 15 minutes

# EXERCISE 3-A

## Expression of the Zero After Adding the Resistor




- $\frac{V_i}{R+\frac{1}{s_z * C_c}} = g_{m2} * V_i$
- $s_z = \frac{1}{C_c * (\frac{1}{g_{m2}} - R)}$

# EXERCISE 3-A

## Finding Value of $R$

- $\omega_{z1} = \omega_{p1} = 15.4 Grad/s = \left| \frac{1}{C_c * \left( \frac{1}{g_{m2}} - R \right)} \right|$
- $R = 64.9\Omega$

# EXERCISE 3-B

## Finding Phase Margin

- $Total\ phase\ shift = 0 - tan^{-1}\frac{2Grad/s}{20Mrad/s} - tan^{-1}\frac{2Grad/s}{18.2Grad/s} = -95.7^o$
- $Phase\ margin = -95.7^o - (-180^o) = 84.3^o$

# FINAL REMARK

## $|p_2|/GBW_T$ vs Phase Margin in Two-poles System

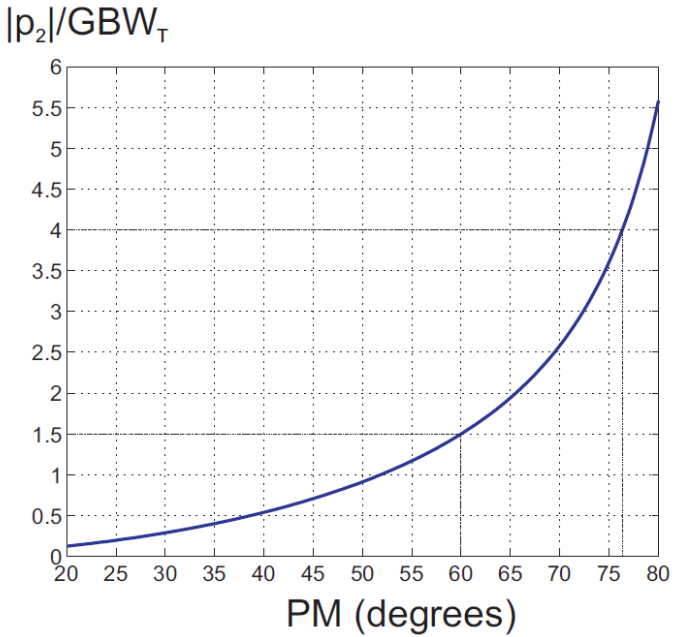



*Figure 7.3: Relation between the phase margin and the ratio of the second pole and $GBW_T$.*




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